Homework 3: Intro & Algorithms (15 points)

Due to Gradescope by 11:45 PM on Tuesday, September 6th
You need to submit a pdf to Gradescope; failure to assign questions to pages may result in a 10% deduction on your grade

Homework Goal: Practice using and critically thinking about algorithms in general and search & sort algorithms in particular; review terminology for these chapters

CS Basics & zyBooks

1. Section 1.2 in zyBooks discusses several historical figures in computer science. Here are two other important people in computer science. Read about them online, then write a short paragraph (3-5 sentences) on each that describes what they have done for the computing revolution. These paragraphs should be in your own words. (2 points)

   (a) Donald Knuth

   Donald Knuth is considered the father of the analysis of algorithms. He is also known for his book series "The Art of Computer Programming". Also the formatting language called "TeX". TeX then became the standard for submitting typeset-ready scientific and mathematical research.

   (b) Jean Bartik

   Jean Bartik was one of the Army's first programmers, she was also one of the first women in computer science. She was later selected to be a programmer for the "Electronic Numerical Integrator and Calculator". After she then trained a team to convert the ENIAC to a stored program machine. In her later years she programmed the "Binary Automatic Computer".

2. Figure 1.4.4 in zyBook provides pictures of several places where embedded devices reside. Provide two other examples where embedded devices exist. (1 point)

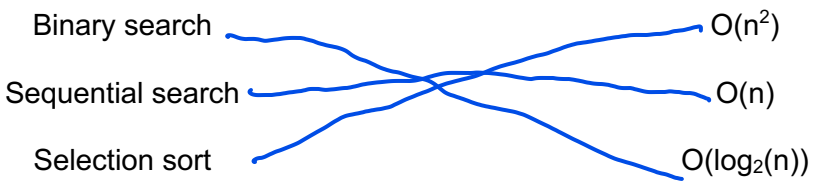
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3. Fill in the blanks: (1 point)

   Computer science is the study of computers, algorithms.
   A(n) algorithm is a set of step-by-step instructions for solving a problem.
   Pseudocode lies between English and machine code.

Big O Notation

4. Match the following algorithms with their time complexity. (1 point)



5. Simplify the following Big O Expressions: (2 points)

   (a) $O(500n + n^2)$ $O(n^2)$

   (b) $O(2 + 6)$ $O(8)$

   (c) $O(n^2 + n*\log(n) + 2^n)$ $O(2^n)$

   (d) $O(n! + n^2)$ $O(n!)$

Searching and Sorting

6. Use the following list to answer the following questions: (2 points)

   [49, 71, 106, 124, 128, 148, 165, 168, 204, 258, 266]

   (a) How many list elements will be compared to find 124 using sequential search?

   4

   (b) How many list elements will be compared to find 124 using binary search?

   3

   (c) How many list elements will be compared to find 204 using linear search?

   9

   (d) How many list elements will be compared to find 204 using binary search?

   2

7. In Fall 2021, Mines had 7,172 students enrolled. Suppose we have a database with everyone's name and phone number at Mines, sorted by full name. In the worst case, how many comparisons would be needed to find the following: (1 point)

   (a) If Paul C. Johnson is a student

   13

   (b) If the phone number 303-273-3000 belongs to a student (no, you can't just call the phone number, you must use the database)

   7,172

8. Given a *large* list of items, should one always sort the list and then use binary search (versus just using sequential search on the large list)? Justify your response. (1 point)

You shouldn't sort because sometimes sorting can take longer than just using sequential search.

Following and Evaluating Algorithms

9. The following is a 2,300-year-old algorithm: (3 points)

Step 1	Get two positive integers as input; call the larger value I and the smaller value J
Step 2	Divide I by J, and call the remainder R
Step 3	If R is not 0, then reset I to the value of J, reset J to the value of R, and return to Step 2
Step 4	Print out the answer, which is the value of J
Step 5	Stop

(a) Go through this algorithm using the input values 18 and 14. Label what step of the algorithm (1-4) you are on. After each step of the algorithm is complete, give the current values of I, J, and R in the table. Determine the final output of the algorithm.

Step #	I	J	R
1	18	14	--
2	18	14	4
3	14	4	4
4	14	4	2
5	4	2	0
6	4	2	0
7	Print J = 2		

(b) What is the use of this algorithm? What does it compute?

Greatest Common Denominator of the two values inputed

(c) Does the algorithm work correctly when the two inputs are 0 and 22? Describe what happens and modify the algorithm to give an appropriate **error message** if the smaller value is 0.

No because dividing by zero is undefined. You should add an error to show why the smaller can't be zero.

10. The following algorithm correctly determines whether summing a large list of random numbers exceeds 10,000. How might you improve the algorithm to make it more efficient? (Hint: adding the numbers one by one is the same as "add all the numbers together") (1 point)

Step 1: Initialize sum = 0.
Step 2: Start with the first number as the sum.
Step 3: Add the next element to the sum.
Step 4: Repeat step 3 until all numbers have been added.
Step 5: Compare the sum with 10,000.

Remove step 5. After step 3 than check the sum. If the program is greater than 10,000 end else continue